

Aditya Tayal

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Final-year CSE undergraduate at IIT Mandi who ships software people actually use. Solo-built PlanMyDegree, the degree planner **1,741 IIT Mandi students** use, plus exam, leave and lab systems for the institute. Five startup internships, **All India Rank 1** in an IEEE ICASSP challenge, 20+ hackathons, Head TA for two core courses, and elected **UG Academic Affairs Secretary** for ~3,500 undergraduates.

EDUCATION

Indian Institute of Technology Mandi

B.Tech in Computer Science & Engineering | CGPA: 8.80

Mandi, Himachal Pradesh

Aug 2023 – May 2027 (expected)

Coursework: Machine Learning, Deep Learning, Generative AI, Data Science, Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks, Information Security, Optimization

Ramakrishna Mission

Class XII, PCM with Computer Science | 89.40%

Gwalior, Madhya Pradesh

2020 – 2022

Wendy School Junior College

Class X, ICSE | 98.20%

Gwalior, Madhya Pradesh

2007 – 2020

EXPERIENCE

Major Technical Project, American Express

Aug 2026 – Present

Imbalanced fraud detection | Python, scikit-learn, XGBoost, LightGBM, CatBoost

Team project

- Stress-testing Self-Paced Ensemble learning on a multi-product credit-card fraud portfolio built from public data.
- Compared Decision Tree, XGBoost, LightGBM and CatBoost base learners, pooled vs per-product models, over 3 seeds.
- Built a cost model that turns a fraud ranking into money saved, reported alongside AUPRC, ROC-AUC, F1 and Recall@K.
- The best configuration recovers **64%** of recoverable fraud loss; findings written up as a full report with tables and figures.

Software Developer Intern, MagoStays

Sep 2026 – Present

Koha Consultants Pvt. Ltd. | TypeScript, Node.js, PostgreSQL, Next.js, React Native

Travel tech

- Full-stack developer on a villa and farmstay booking platform, across the backend, admin panel, website and host app.
- Own payments, invoicing and booking features end to end, from database schema and APIs to the user interface.
- Write unit, integration and end-to-end tests for each feature I ship, plus a detailed design document for every flow I own.
- Work closely with the product and finance teams on edge cases, audit trails and safe, reversible database migrations.

Software Developer Intern, InstaFarms

Apr 2026 – Aug 2026

VWA Hospitality Pvt. Ltd. | TypeScript, Node.js, PostgreSQL, Next.js

Travel tech

- Full-stack developer on a farmstay and villa booking platform, working across the backend, admin panel and website.
- Built booking, pricing and tax features end to end in a TypeScript monorepo with typed, self-documenting APIs.
- Integrated a channel partner's inventory into checkout, and sped up the availability calendar with batched API calls.
- Added a new property type across every app, and improved the website's search, SEO pages and page rendering.

AI/ML Engineer Intern, WattCycle

Dec 2025 – Jun 2026

Solar energy | Python, FastAPI, scikit-learn, XGBoost, Prophet, PostgreSQL, React

- Worked across ML and full stack on WattCycle Atlas, a monitoring and forecasting platform for solar power plants.
- Built fault detection pairing rule-based plant health checks with Isolation Forest anomaly scores and severities.
- Designed multi-horizon energy forecasting with XGBoost and Prophet models, served to the dashboards via FastAPI.
- Built the analytics backend behind its dashboards and containerised the services with Docker and CI pipelines.

Full Stack Developer Intern, Garudakshak Technologies

Oct 2025 – Apr 2026

Defence tech | Tauri, Go, React, TypeScript, PostgreSQL

- Built the operator desktop application for a counter-drone system, taking it from an empty repository to an MVP.
- Cross-platform Tauri app with a Go backend and a React + TypeScript interface, shipped for Windows and Raspberry Pi.
- Real-time operator dashboards over WebSockets, with role-based access control and automated release builds in CI.
- Worked closely with the hardware and ML teams to bring their detection models into a single operator interface.

Lead Developer Intern, Strive Partners

Sep 2025 – Jun 2026

Hiring tech | React, TypeScript, JavaScript, Django REST, AWS

- Led development of Strive Hunt, a hiring platform connecting candidates, employers and an in-house recruiting team.
- Built the candidate and employer portals, and an internal applicant-tracking system used daily by the recruiters.
- Real-time shared candidate sheets, bulk Excel import with column mapping, and candidate search with rich filters.
- Secured the app with token refresh, CSRF protection and input sanitising; shipped through Docker and CI pipelines.

PlanMyDegree, Academic Planner (Solo)*Next.js, TypeScript, Prisma, PostgreSQL, NextAuth, PWA**Jan 2026 – Present*

- Solo-built the degree planner **1,741 IIT Mandi students** use to plan credits, minors and timetables each term.
- **150+ sign-ups in 36 hours** from a single WhatsApp message, and 17,555 course enrollments tracked since launch.
- Pre-registration with seat caps and course exclusions, timetable clash detection, transcript OCR and calendar export.
- 30 data models, ~65 API routes and 880+ commits, shipped as an installable PWA rated 4.9/5 in student feedback.

Camouflaged Object Detection (Paid Research)*PyTorch, mmdetection, DINOv2, LoRA, SigLIP-2, SAM2**Apr 2026 – Jul 2026*

- Adapted PAIS (ICCV 2023) to camouflaged objects with only 10% labelled data: **+7.8 AP** over the supervised model.
- Designed a text-guided DINOv2 + LoRA decoder with SigLIP-2 queries: S-measure **0.847** on COD10K, 0.865 on NC4K.
- Reproduced UCOD-DPL (CVPR 2025) with two-pass SAM2 refinement and soft voting: **0.8407 vs the paper's 0.835**.
- Beat the reproduced GBNet baseline on **42/42** dataset-metric checks across 6 datasets and 71 documented experiments.

Domain Unlearning in CLIP (Team of 4)[Code](#)*PyTorch, CLIP ViT-B/16, prompt learning, SCRUB**Jan 2026 – May 2026*

- Made CLIP forget a whole visual domain but keep the rest, beating the published result (**72.10 vs 69.96 H-score**).
- Found the forget signal is outvoted N:1 by the domains being kept, and rebalanced the loss so forgetting works.
- Adapted SCRUB from forgetting single samples to forgetting whole domains: **83.6 H-score**, **+13.6** over the paper.
- Extended it to forgetting two domains at once (82.9 average over 6 pairs) and wrote it up as an IEEE-format report.

ALS Severity from Voice, SAND 2025 (Team of 4)*Python, audio machine learning**Oct 2025 – Nov 2025*

- Graded ALS severity from patients' speech for an IEEE ICASSP 2026 challenge: **All India Rank 1, Global Rank 11**.
- Scored macro-F1 0.5430 on the 5-class severity task, and ranked India #2 (Global 19) on the disease-progression task.

Exodia 2026 Ticketing & Gate System (Solo)[Code](#)*Next.js, tRPC, Prisma, PostgreSQL, Nodemailer**Mar 2026 – Apr 2026*

- Built QR ticketing, bulk mailing and gate scanning alone for the fest's biggest night: **300+ tickets, 200 entries**.
- Atomic single-use database claims stopped **26 reuse attempts**, so **0 duplicate entries** got through the gate.
- A throttled multi-account mailer with open tracking; **13 commits** shipped live on event day with zero downtime.

Faculty Leave Portal (Solo)*Fastify, Prisma, React, SQLite, Docker**Sep 2026 – Present*

- Leave management for IIT Mandi faculty, built from the institute's own requirements document and its review.
- No stored balances: every figure is derived from an append-only ledger that the database itself refuses to edit or delete.
- **324 requirements** traced to **321 automated tests**, with email OTP sign-in, CSRF guards and file-upload validation.

TestPlatform, Online Examinations (Solo)*Next.js, Express, Socket.IO, PostgreSQL, Redis**Aug 2026 – Present*

- An exam system where the clock lives on the server, not the browser: Redis-held timers, live rooms, pause and resume.
- Differently shuffled papers per student, auto-grading for MCQ, multi-select and numeric answers, passwordless OTP.
- Hardened after a self-audit: row-locked submissions, IDOR fixes, a pinned JWT algorithm and patched dependency CVEs.

Vehicle Counting with Deep Learning Banned (Solo)*Python, OpenCV, NumPy**Jan 2026*

- Counted traffic with classical computer vision only: a 9-stage pipeline built in 48 hours, **rank 10 against teams**.
- Background subtraction, centroid tracking and track merging with a night mode; **MAE 16.4** on the hidden test set.

MahaDrishti AI, Satellite Crop Verification (Team of 3)*Sentinel-1/2, Prithvi-EO, Random Forest, FastAPI**May 2026*

- Checks a farmer's declared crop against Sentinel-1 and Sentinel-2 imagery plot by plot, with reports in Marathi.
- **Finalist** at the Pune International Agri Hackathon; processes all 194 plots of a village in under 2 minutes end to end.

Product Price Prediction, Amazon ML Challenge (Team of 3)*Python, LightGBM, TF-IDF**Oct 2025*

- Priced 75,000 products from their text alone with a 21-model LightGBM ensemble trained over 7 folds and 3 seeds.
- Cut SMAPE from **54.38% to 32.33%** out-of-fold using TF-IDF text features, regex-mined numbers and calibration.

FinFabric, IDBI Innovate 2026 (Team Lead)[Code](#)*JavaScript, scoring models, Vercel**Jul 2026*

- Led a team building 5 live banking prototypes: MSME health scores, SIP planning, lead scoring and default alerts.
- Each one shipped as its own live web app with a working scoring engine, plus a fuzzy-matching reconciliation copilot.

Sahayak, KSP Datathon 2026 (Solo)[Code](#)*Qwen 2.5 Coder, FastAPI, React**Jun 2026*

- Ask a crime database questions in English or Kannada and get back the SQL, the source rows and matching legal codes.
- A read-only SQL guard means a generated query can never change data; served by Qwen 2.5 Coder behind FastAPI.

Deepfake Detection (Team of 6)

CLIP, DINOv2, SVM (cuML)

Feb 2025 – May 2025

- Caught AI-generated images from **19 generators** with frozen CLIP and DINOv2 features and an RBF-kernel GPU SVM.
- **89.4%** weighted accuracy across 80,353 test images, without fine-tuning the frozen backbones on any generator.

CaptureSmart, AI Camera Assistant (Team of 6)

Kotlin, CameraX, TensorFlow, Flask

[Code](#)

May 2025

- An Android camera that scores motion blur live with MobileNetV2 (Pearson **0.89**) and suggests ISO and shutter speed.
- Trained on 6,000 hand-collected and 18,000 synthetic images; I built the app and its Flask model-serving layer.
- Built in a 48-hour IIT Mandi hackathon and placed **3rd of 60+ teams**; two small MLPs predict the camera settings.

Adobe India Hackathon 2025 (Team Leader)

Python, PyMuPDF, MiniLM, Docker

[1A](#) [1B](#)

Jul 2025

- **Top 1.5%** of 262,120 teams: multilingual PDF outline extraction in a fast, offline, CPU-only Docker container.
- Persona-aware section ranking with MiniLM embeddings that processes 7 PDFs in under 45 seconds on a plain CPU.

DeepMatch, India Runs Hackathon (Solo)

Python, sentence-transformers

[Code](#)

Jun 2026

- Ranks 100K job candidates to a shortlist in **~75 s on a CPU**: rules, then MiniLM scoring, then signal fusion.
- Six honeypot heuristics flag fake or padded profiles before ranking, and peak memory stays near 3 GB on a laptop.

SevaVoice, HACKHAZARDS '26 (Solo)

React, TypeScript, Express, Sarvam AI

[Live](#) [Code](#)

Jun 2026

- Turns confusing government and public-service notices into verified, step-by-step action plans in **14 Indian languages**.
- Reads the plans aloud in 11 languages via Sarvam speech APIs, with budget and rate guards and 63 Playwright checks.

DyslexoFly, Code for Bharat S2 (Team of 2)

Next.js, Flask, DistilBART, mT5, edge-tts

[Live](#) [Code](#)

Jun 2025 – Jul 2025

- Turns PDFs and documents into dyslexia-friendly reading with OpenDyslexic, spacing controls and 3-level AI summaries.
- Hindi and English summaries plus audio in 7 neural voices; reached the top 100 of 1,800+ Code for Bharat entries.

IIT Mandi LaTeX (Lead)

Overleaf CE, Node.js, Docker

Sep 2026 – Present

- An institute-hosted Overleaf for ~3,500 users with a fair first-come compile queue and a 50-user waiting room.
- Per-user storage quotas, nightly off-host backups and an IIT Mandi template pack, built on Overleaf Community Edition.

TinkerBench & Simulation Lab (Solo)

JavaScript, Canvas, nginix

Sep 2026 – Oct 2026

- A browser electronics bench that solves circuits by nodal analysis every frame and runs Arduino code in the page.
- **49** interactive physics and chemistry experiments for school labs, each laid out as a live lab record with graphs.

Faculty Application Portal

Django, SQL

[Live](#)

Dec 2024 – Feb 2025

- Replaced Google Forms with a Django portal on IIT Mandi's recruitment site, with an admin dashboard for shortlisting.
- Automated CV generation and applicant emails, and added filtering, tracking and export for the recruitment office.

More projects

- **Face EngagementCV** [Code](#): names every face in classroom CCTV and scores engagement from brightness, eyes and motion.
- **PSCDL 2026** [Code](#): persistent scene-change detection with classical CV, two persistence clocks and adaptive thresholds.
- **Vidyut AI** [Code](#): cited answers to industrial-safety questions from OSHA, CSB and Indian standards, in Hindi too.
- **FlowAI** [Code](#): a Gemini multi-agent planner that splits goals into subtasks, schedules them and sends you reminders.
- **PurrPedia** [Code](#): a 247-breed cat encyclopedia with Temporal-scheduled postcards; all 5 security-audit findings fixed.
- **CareerAutofill AI** [Code](#): planner agents and a Chrome extension that fill job forms; LoRA-tuned distilgpt2 drafts answers.
- **TheBudgetBuddy** [Code](#): plain-English questions over your own transactions, via LangChain, Llama3-70B and FAISS.
- **SecureInvest-AI** [Code](#): a fraud-risk scorer for investment pitches, built solo in 48 hours for the SEBI hackathon.
- **Malware scanner** [Code](#): static .exe, .pdf and .docx scanning with YARA rules, PE import analysis and macro checks.
- **TheAttendioBot** [Code](#): a Telegram bot that tracks attendance, plans safe skips and sends reminders, on Google Sheets.
- **Neufin** [Live](#) [Code](#): a product catalogue with search, pagination and a cart, validated with Zod and tested with Jest.
- **CyberShield**: mule-account detection on 9,082 bank accounts at a 0.89% base rate, PSBs Hackathon at IIT Hyderabad.
- **PodSense**: a four-agent Kubernetes pod observability design with a central orchestrator, for ABB Accelerator 2026.
- **Solar-panel waste policy**: tested whether battery-recycling rules can carry over to solar waste, with a plant visit.
- **PRAYAS ML labs**: four teaching notebooks, from decision trees to DQN and TFLite, with 60 guided experiments.
- **Campus self-hosting**: six campus services on one Raspberry Pi 5, reached through self-updating GitHub redirects.
- **Tele-operated ROS robot**: a Wi-Fi controlled robot on a custom PCB that retraces its own path, for a design practicum.
- Also: a research-group website, the GCS cell website, a HackerRank Orchestrate entry and LaTeX notes for core courses.

ACHIEVEMENTS

AIR 1, Global Rank 11, SAND 2025 (IEEE ICASSP 2026 grand challenge): ALS severity from voice; India #2 in Task 2.
Rank 10, solo against teams, Vehant Vehicle Count Challenge 2026: classical computer vision only, MAE 16.4 in 48 hours.
Finalist, Pune International Agri Hackathon 2026 (Govt. of Maharashtra and MPKV), geospatial crop survey track.
3rd place, AiHack India 2025, hosted at IIT Mandi by Aiful Corporation, Japan; also 3rd of 60+ with CaptureSmart.
Top 1.5% of 262,120 teams, Adobe India Hackathon 2025, as team leader; DyslexoFly in Code for Bharat's top 100.
Emerald of Gymkhana Award 2025: best GCS Cell volunteer, and Head Volunteer of the 2024 induction programme.
National Semi-Finalist, Tata Imagination Challenge 2024, through two national rounds of strategy and critical thinking.
1st place, SpaceCode Challenge 2023: a 48-hour team hackathon in computational astrophysics, run by STAC, IIT Mandi.
Code2Career placement test 2025 (Let's Code): ranked 128 in coding, 148 in aptitude and 182 in CS fundamentals.
20+ hackathons in 2025–26, incl. Amazon ML Challenge, IDBI Innovate (team lead), ABB Accelerator and SEBI GFF.

TEACHING

B.Tech Head Teaching Assistant & Teaching Assistant, IIT Mandi *Aug 2025 – May 2026*
– Head TA for Machine Learning (Aug–Dec 2025) and Data Structures & Algorithms (Jan–May 2026), under Dr. Indu Joshi.
– Ran course operations for **600+ students**: labs, assignments, evaluation, invigilation and weekly doubt-clearing sessions.
– TA for Introduction to Python & Data Science (Aug–Dec 2025) and for Probability & Statistics (Jan–May 2026).
Lab Lead, PRAYAS 4.0 & HIMSHIKHAR 2026, CCE IIT Mandi *Jun 2026 – Jul 2026*
– Lab lead for AI/ML/DL at PRAYAS 4.0 and Software Development & AI at HIMSHIKHAR: 22 classes, 40+ hours.
– Taught learners from Class 9 to over 80, from DBMS to GenAI, and led **8 TAs** through 4 hands-on Colab labs.
Teaching Assistant, Masai (Core Computer Science) *Jun 2025 – Present*
– TA for a core computer-science module covering database systems, system design, operating systems and network security.
– Resolve doubts and mentor working learners alongside the instructional team, through live sessions and reviews.

POSITIONS OF RESPONSIBILITY

UG Academic Affairs Secretary, Student Gymkhana, IIT Mandi *Jul 2026 – Present*
– Elected student lead for academics for ~**3,500 undergraduates**; head the UG Academic Council and its councillors.
– Picked **12 branch councillors** and ran Batch 2026's academic orientation across **15 branch-wise sessions** in August.
Academic Councillor & Sub-Councillor, Batch 2023, UG Academic Council *Feb 2024 – Jul 2026*
– Sub-Councillor (2024–25), then Councillor (2025–26) and batch representative for **500+ students** of Batch 2023.
– Co-ran the first post-COVID offline curriculum session for freshers and hired the council's web team through a live build.
Guidance & Counselling Services, IIT Mandi: Head Volunteer, Coordinator, Mentor *Jun 2024 – Present*
– Head Volunteer for the 2024 induction programme, leading volunteers who counselled and settled in the incoming batch.
– Coordinator of the counselling cell for 2025–26, then Mentor from 2026; won the Emerald of Gymkhana for best volunteer.
SAE Club Co-Coordinator & Mentor; STAC Core Team Member *Aug 2024 – Present*
– Co-Coordinator of the SAE club (2024–25), running its sessions and team activities, and its Mentor since July 2025.
– Core team of the Space Technology & Astronomy Cell (2024–25), co-running astronomy hackathons for 100+ teams.
Head of Technical Team, Exodia 2026 (Annual Cultural Fest) *Apr 2026*
– Ran the fest's tech, from the ticketing and gate system to code splitting, image and loading work on its website.
Hospitality Head & Orion Hackathon Organiser, Xpecto 2025 (Technical Fest) *Nov 2024 – Apr 2025*
– Led hospitality for IIT Mandi's technical fest and its visiting teams, and co-organised the Orion hackathon at AstraX'25.
JEE Mentor, Physics Wallah *May 2024 – Present*
– Mentored **100+ JEE aspirants** through 1:1 sessions on preparation strategy, problem solving and exam temperament.
Volunteering: National Service Scheme & Campus Events, IIT Mandi *Sep 2023 – Present*
– NSS member since 2023; volunteered at Convocation 2024, Diwali 2023 and 2024, and Janmashtami 2023 on campus.

CERTIFICATIONS

AI/ML for Geodata Analysis, Indian Institute of Remote Sensing (ISRO), 2024; **Data Analytics Essentials**, Cisco, 2024.
Software Engineering Job Simulation, Goldman Sachs on Forage, 2024; **Flipkart GRiD 6.0**, Level 1, E-commerce, 2024.
IBM: Data Fundamentals, Getting Started with Data and with AI, 2024; **Git and GitHub**, Physics Wallah, 2025.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Go, C++, C, Java, Kotlin, SQL, Rust, R, HTML/CSS, LaTeX, Bash
Backend: FastAPI, Django, Flask, Node.js, Express, Fastify, Hono, tRPC, REST APIs, WebSockets, Socket.IO
Frontend & Mobile: React, Next.js, Tailwind CSS, Vite, Astro, Tauri, React Native, Expo, Android (Kotlin), PWAs
ML & AI: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, OpenCV, Hugging Face, LoRA, RAG, LangChain
ML Areas: Machine Learning, Deep Learning, Computer Vision, NLP, LLMs, Time-Series Forecasting, Anomaly Detection
Data & Cloud: PostgreSQL, Redis, MongoDB, SQLite, Prisma, Docker, AWS, Vercel, Firebase, Linux, Git, GitHub Actions